Assessment of Novelty of DC Solutions

* Required

Information

We are collecting experts' evaluations of the novelty of MV and LV DC technologies/solutions as generated by the large language models (LLMs). The LLMs were prompted to generate novel DC solutions as part of an initial exploration of their capabilities to contribution in expert studies.

1. Continue to the form? *

- ◯ Yes
- ◯ No

Demographics

This part aims to collect participants' demographic information

2. Your domain expertise in DC.
(eg: DC power electronics engineer, electrical engineer, DC protections , DC interoperability, Data center electrification engineer, etc) *

3. Which part of the demonstration site that best describes your expertise experience? *

- ◯ Ports
- ◯ Industry
- ◯ Data Centres
- ◯ Buildings

4. Years of experience *

The value must be a number

Solution 1: Electric Vehicles (EVs) charging stations

This DC solution includes: EV Charging Stations for Industry | EV Charging Infrastructure | Smart Charging Infrastructure for Electric Vehicles | EV Charging Stations for Building Parking Lots | Smart Charging System for EVs | Smart charging infrastructure.

Objective: To promote sustainable transportation by deploying intelligent, renewable-powered electric vehicle (EV) charging systems across industry and buildings.

Description: The solutions involve installing EV charging stations in industrial sites and building parking lots, forming a network powered by solar and wind energy. Artificial intelligence and machine learning optimize charging schedules based on demand, occupancy, weather, and traffic patterns. Together, these systems enable efficient energy use, seamless charging experiences, and reduced reliance on fossil fuels.

5. Based on this description, can you judge the novelty of the EV charging stations? *

- ◯ Yes
- ◯ No

6. How novel are the EV charging stations in your field of expertise? *

	Zero/no novelty	Low novelty	Medium novelty	High novelty	Very high novelty
Novelty	◯	◯	◯	◯	◯

7. Rate the EV charging station in comparison to the most comparable existing DC solution you are aware of. *

	Nearly identical	Minor variations	Moderately different	Substantially different	Fundamentally different
Similarity	◯	◯	◯	◯	◯

8. Which aspect drive your novelty judgment? *

- ◯ New combination of known components
- ◯ New mechanism/technical principle
- ◯ New application/context for known technologies
- ◯ Known mechanisms and combination

9. Have you previously seen or worked on a substantially similar DC solution? *

- ◯ No
- ◯ Unsure
- ◯ Yes - in literature
- ◯ Yes - in practice
- ◯ Yes - both in practice and literature

10. How confident are you in this novelty judgment? *

	Not at all confident	Somewhat different	Neutral	Confident	Very confident
Confidence interval	◯	◯	◯	◯	◯

11. Reason for selecting 'no' *

12. Any comments on the proposed EV charging stations?

Solution 2: Intelligent Building Energy Management System (IBEMS)

This DC solution includes: Building Management System (BMS) | Energy Management Systems for Buildings (EMS) | Intelligent Building Management System | Building Automation Systems | Building Integrated Photovoltaic Systems (BIPV) | Integrated Renewable Energy Management System.

Objective: To optimize building energy consumption and enable on-site renewable energy generation through the integrated management of intelligent control systems and building-integrated photovoltaic technologies.

Description: The Intelligent Building Energy Management System (IBEMS) integrates advanced energy management and automation with building-integrated photovoltaics (BIPV) and renewable energy management to optimize energy use and generate on-site DC electricity. It combines real-time monitoring, intelligent controls, and integrated renewable generation or storage to improve energy efficiency, self-sufficiency, and sustainability while reducing costs and emissions in buildings.

13. Based on this description, can you judge the novelty of the IBEMS? *

- ◯ Yes
- ◯ No

14. How novel are the IBEMS in your field of expertise? *

	Zero/no novelty	Low novelty	Medium novelty	High novelty	Very high novelty
Novelty	◯	◯	◯	◯	◯

15. Rate the IBEMS in comparison to the most comparable existing DC solution you are aware of. *

	Nearly identical	Minor variations	Moderately different	Substantially different	Fundamentally different
Similarity	◯	◯	◯	◯	◯

16. Which aspect drive your novelty judgment for IBEMS? *

- ◯ New combination of known components
- ◯ New mechanism/technical principle
- ◯ New application/context for known technologies
- ◯ Known mechanisms and combination

17. Have you previously seen or worked on a substantially similar DC solution? *

- ◯ No
- ◯ Unsure
- ◯ Yes - in literature
- ◯ Yes - in practice
- ◯ Yes - both in practice and literature

18. How confident are you in this novelty judgment? *

	No at all confident	Somewhat confident	Neutral	Confident	Very confident
Confidence interval	◯	◯	◯	◯	◯

19. Reason for selecting 'no' *

20. Any comment(s) on the proposed intelligent building energy management systems (IBEMS)?

Solution 3: Solar PV Tracking System

This DC solution includes: Solar PV Tracking System | Solar Tracking System | Solar PV Tracking | Solar PV Tracking System for Ports | Solar PV Tracker | Solar-Powered Ports.
Objective: To maximize solar energy generation and support sustainable operations through photovoltaic power systems and intelligent tracking or orientation technologies.
Description: The solutions combine solar photovoltaic power deployment with advanced tracking and orientation systems to increase renewable electricity generation. They include solar installations for ports and related infrastructure, alongside sensors, algorithms, and control technologies that adjust panel positioning based on the sun's movement, weather, and seasonal conditions. Together, these systems improve energy yield, support cleaner operations, and reduce reliance on fossil-fuel-based power.

21. Based on this description, can you judge the novelty of the solar PV tracking system? *

- ◯ Yes
- ◯ No

22. How novel are the solar PV tracking systems in your field of expertise? *

	Zero/no novelty	Low novelty	Medium novelty	High novelty	Very high novelty
Novelty	◯	◯	◯	◯	◯

23. Rate the solar PV tracking systems in comparison to the most comparable existing DC solution you are aware of. *

	Nearly identical	Minor variations	Moderately different	Substantially different	Fundamentally different
Similarity	◯	◯	◯	◯	◯

24. Which aspect drive your novelty judgment for solar PV tracking system? *

- ◯ New combination of known components
- ◯ New mechanism/technical principle
- ◯ New application/context for known technologies
- ◯ Known mechanisms and combination

25. Have you previously seen or worked on a substantially similar DC solution? *

- ◯ No
- ◯ Unsure
- ◯ Yes - in literature
- ◯ Yes - in practice
- ◯ Yes - both in practice and literature

26. How confident are you in this novelty judgment? *

	Not at all confident	Somewhat confident	Neutral	Confident	Very confident
Confidence interval	◯	◯	◯	◯	◯

27. Reason for selecting 'no' *

28. Any comment(s) on the proposed solar PV tracking system?

Solution 4: Battery Energy Storage Systems

This DC solution includes: Energy Storage Hybrid System | Solar Hybrid System | Solar Hybrid System for Ports | Battery Energy Storage System | Solar PV Hybrid Power System | Solar PV and Battery Storage | Fuel Cell and Battery Storage | Energy Storage Systems for Buildings.
Objective: To ensure a stable and reliable power supply by integrating renewable energy sources with efficient storage systems.
Description: The hybrid solutions combine solar, wind, hydro, or fuel cell technologies with batteries or conventional generators to balance energy demand and supply. By storing excess renewable energy and releasing it during low-output or peak-demand periods, these systems enhance reliability, resilience, and energy efficiency across ports, industry, data centers, and buildings—reducing dependence on non-renewable sources while optimizing sustainability.

29. Based on this description, can you judge the novelty of the battery energy storage systems? *

- ◯ Yes
- ◯ No

30. How novel are the battery energy storage systems in your field of expertise? *

	Zero/no novelty	Low novelty	Medium novelty	High novelty	Very high novelty
Novelty	◯	◯	◯	◯	◯

31. Rate the battery energy storage systems in comparison to the most comparable existing DC solution you are aware of. *

	Nearly identical	Minor variation	Moderately different	Substantially different	Fundamentally different
Similarity	◯	◯	◯	◯	◯

32. Which aspect drive your novelty judgment for battery energy storage systems? *

- ◯ New combination of known components
- ◯ New mechanism/technical principle
- ◯ New application/context for known technologies
- ◯ Known mechanisms and combination

33. Have you previously seen or worked on a substantially similar DC solution? *

- ◯ No
- ◯ Unsure
- ◯ Yes - in literature
- ◯ Yes - in practice
- ◯ Yes - both in practice and literature

34. How confident are you in this novelty judgment? *

	Not at all confident	Somewhat confident	Neutral	Confident	Very confident
Confidence interval	◯	◯	◯	◯	◯

35. Reason for selecting 'no' *

36. Any comment(s) on the proposed battery energy storage systems?

Solution 5: LED Lighting Systems

This DC solution includes: Energy Efficient Lighting for Industry | Energy Efficiency Measures | Industrial Energy Efficiency | Efficient Lighting System for Data Centers | LED Lighting Retrofit | Efficient Lighting Systems in Industry | Energy-Efficient Lighting | High-Efficiency LED Lighting System.
Objective: To improve energy efficiency and reduce operational costs through advanced LED lighting and smart control systems.
Description: The solutions involve efforts to substitute conventional lighting with energy-efficient LED systems across ports, industry, buildings, and data centers. Integrated with smart sensors and intelligent controls, these systems optimize lighting levels and usage based on occupancy and natural light. They consume less electricity, have longer lifespans, and lower maintenance costs while maintaining high-quality illumination and supporting sustainable energy management.

37. Based on this description, can you judge the novelty of the LED lighting systems? *

- ◯ Yes
- ◯ No

38. How novel are the LED lighting systems in your field of expertise? *

	Zero/no novelty	Low novelty	Medium novelty	High novelty	Very high novelty
Novelty	◯	◯	◯	◯	◯

39. Rate the LED lighting systems in comparison to the most comparable existing DC solution you are aware of. *

	Nearly identical	Minor variations	Moderately different	Substantially different	Fundamentally different
Similarity	◯	◯	◯	◯	◯

40. Which aspect drive your novelty judgment for LED lighting systems? *

- ◯ New combination of known components
- ◯ New mechanism/technical principle
- ◯ New application/context for known technologies
- ◯ Known mechanisms and combination

41. Have you previously seen or worked on a substantially similar DC solution? *

- ◯ No
- ◯ Unsure
- ◯ Yes - in literature
- ◯ Yes - in practice
- ◯ Yes - both in practice and literature

42. How confident are you in this novelty judgment? *

	Not at all confident	Somewhat confident	Neutral	Confident	Very confident
Confidence interval	◯	◯	◯	◯	◯

43. Reason for selecting 'no' *

44. Any comment(s) on the proposed LED lighting systems?

Thank you for taking the time to share your expertise.

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